



Origination Models

GINI Effect on Bad Rate/ Losses



EDA Research on Relationship between Gini and Losses for Origination Models

■ The Business Case:

- ▶ The Gini Coefficient is negatively correlated with the bad rate of the accepts in a given population – the higher the Gini the lower the bad rate of accepts
- ▶ Bad Rate of Accepts can indeed be used as a proxy for losses (assuming that Losses are 100% of Exposure for Bad Accounts) so a decrease in the Bad Rate can be regarded as a decrease in losses
- ▶ Additionally it is expected that this effect is positively correlated to the quality of the target population and the conservatism of the bank strategy – the higher the quality of the target population* (lower bad rate of the population) and/or the more conservative the bank (lower acceptance rate), the higher the effect of the Gini increase

■ Research Assumptions:

- ▶ The cumulative distribution of the goods and bads in the population follows a distribution which is a variant of the normal distribution and thus can be described by its Gini-coefficient and the population bad rate (a distribution function was assumed**)
- ▶ The analysis is implemented for different levels of population credit quality (measured as a target population bad rate) and different accept rates – the graphs in the presentation include only two examples of the carried out analysis.

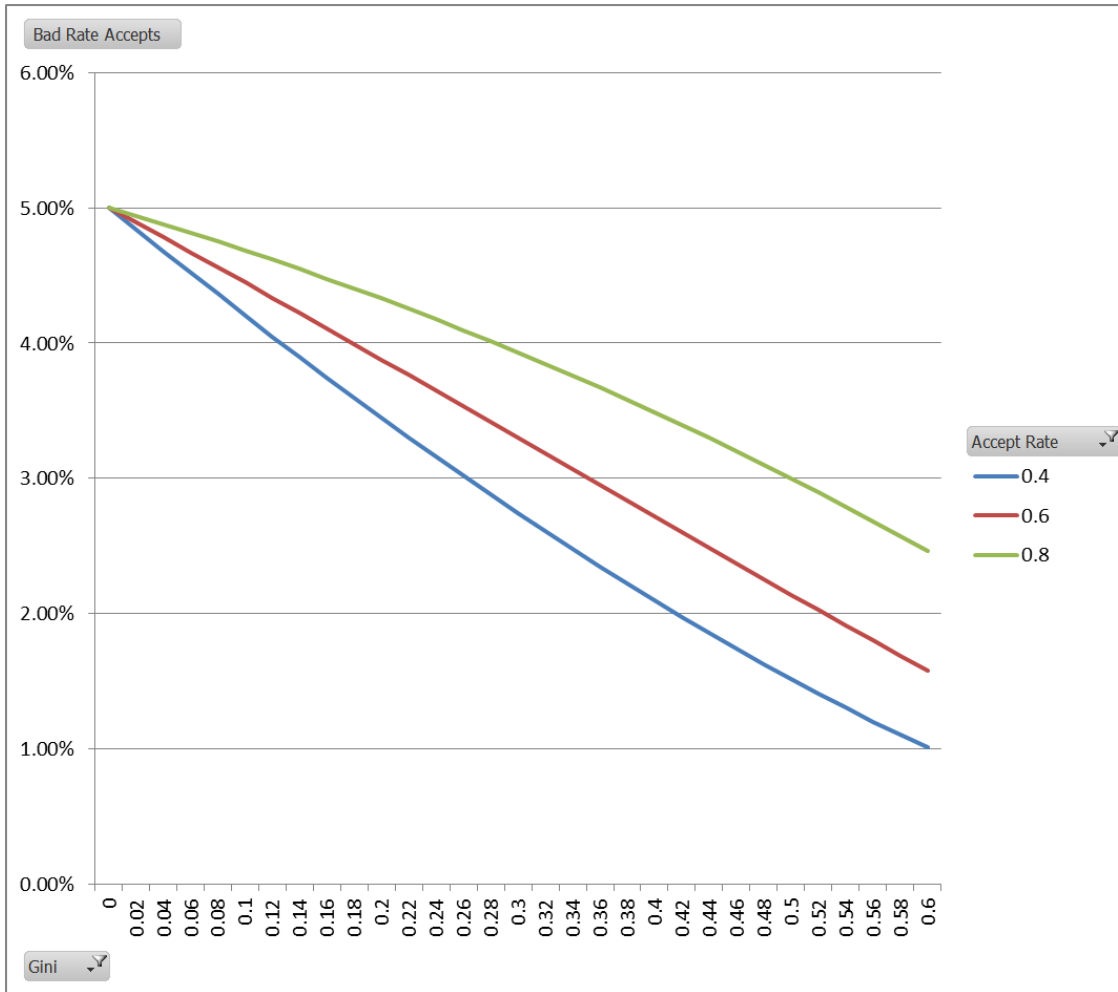
*Target Population is the whole population scored with the model (this is the population of applicants not hitting the policy reject rules)

** The Lorenz curve parameterization gives the following links:

$$\%G = \Phi(\Phi^{-1}(\%B) + \sigma) \quad \text{and} \quad \sigma = \sqrt{2} \cdot \Phi^{-1}((\text{Gini} + 1)/2)$$



Bad Rate of Accepts decrease with increasing Gini at the same Acceptance Rate (Volume of Business)

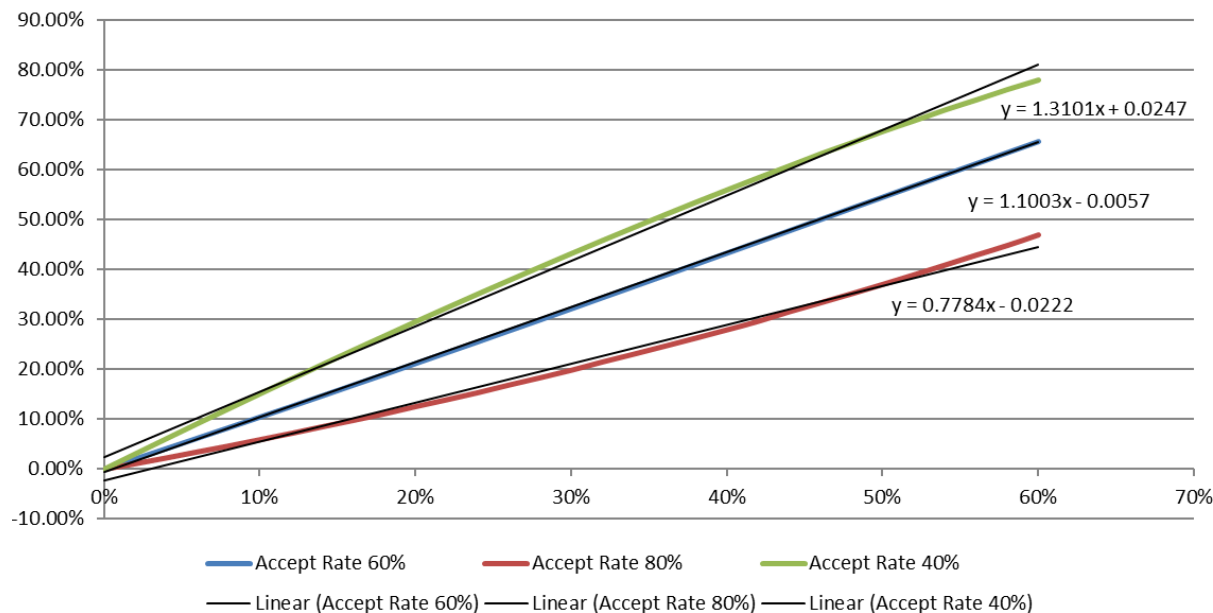


- Graph – The calculated bad rates of Accepts (y-axis) in relation to Gini (x-axis) for three different acceptance rates (0.4, 0.6 and 0.8 of the entire population) and a population bad rate of 5%
- The higher the Gini, the better the quality of the accepted portfolio (lower bad rate of accepts) and therefore the lower the losses
- The more conservative the strategy (lower accept rate), the stronger the increase of portfolio quality with increasing Gini as suggested by the curves' shape



Relationship between GINI and Bad Rate of Accepts for Different Acceptance Rate Targets

Decrease in % Bad Rate of Accepts, compared to Gini = 0



- Graph – The decrease of the accepts' bad rate (as compared to a Gini of 0%), as a function of the Gini (x-axis)
- The Graph is for Population Bad Rate of 10%
- The blue curve shows the decrease in the bad rate with an almost linear trend – an increase in Gini of 10% would lead to a decrease in the bad rate of accepts of appr. 11%

- The lower the general accept rate and the lower the population bad rate – the steeper the curve – for Accept Rate of 40% a GINI increase of 10% would yield a Bad Rate decrease of 13%, for Accept Rate of 80% a GINI increase of 10% would yield a Bad Rate decrease of 8%



Conclusions

- For an Acceptance Rate of 60% it is expected that an Increase in GINI of 10% would yield a similar decrease in Accepts Bad Rate (10-11%) and therefore a similar decrease in losses as compared to a random model (GINI=0%)
- For a higher Acceptance Rate (80%) it is expected that a GINI increase of 10% would yield a lower decrease in Accepts Bad Rate and losses (about 8% for a Target Population* with 10% Bad Rate) as compared to a random model (GINI=0%)
- For a lower Acceptance Rate (40%) it is expected that a GINI increase of 10% would yield a higher decrease in Accepts Bad Rate and losses (about 13% for a Target Population* with 10% Bad Rate) as compared to a random model (GINI=0%)

*Target Population is the whole population scored with the model (this is the population of applicants not hitting the policy reject rules)



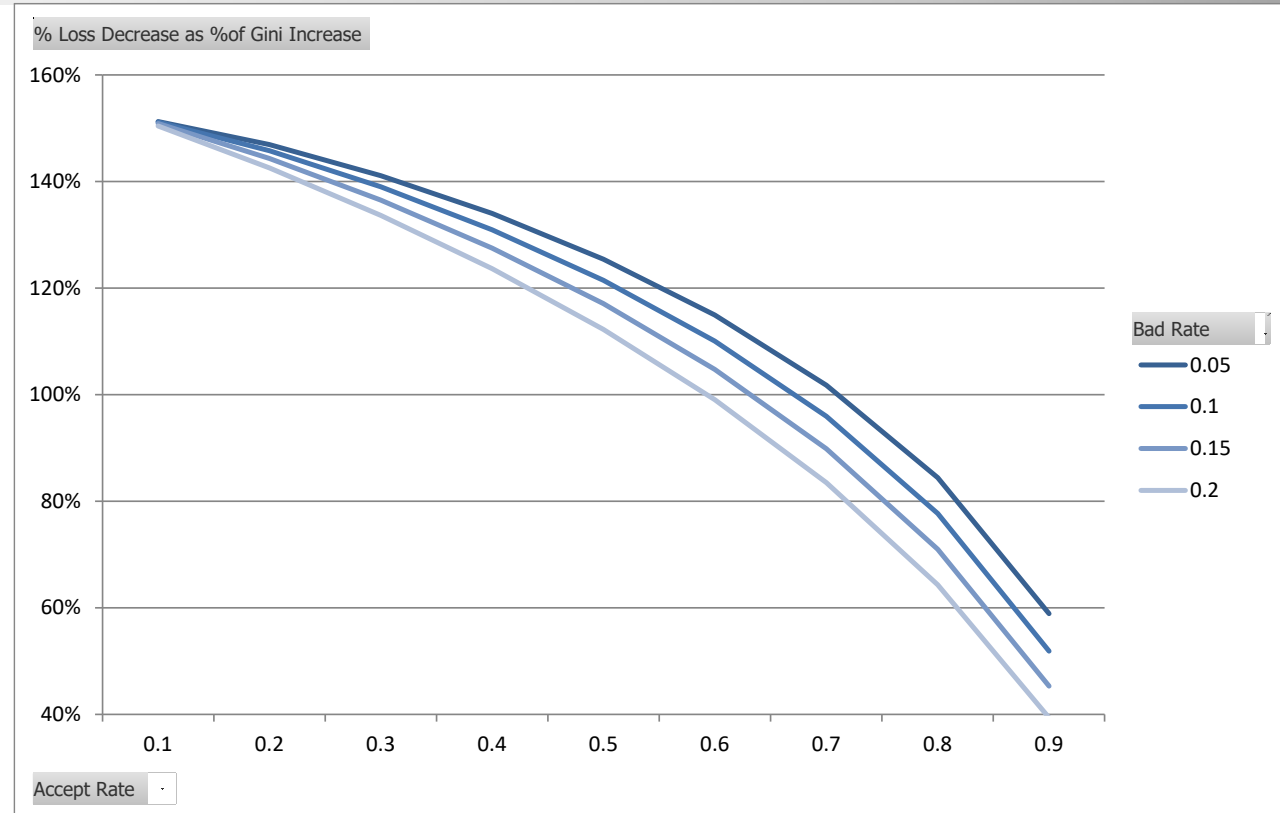
Acceptance Rate effect on the Relationship between Gini and %Loss Decrease

- To measure the relationship between an institution's Acceptance Rate and the effect of Gini on %Losses additional research was done
- The slope of the function that expresses %Loss Decrease for a given Gini (as depicted on slide 3) was calculated for different Population Bad Rates
- The slope was depicted against the Acceptance Rate to illustrate the effect the Acceptance Rate has on %Loss Decrease as a function of the Gini
- The slope was expressed in % value which could be interpreted as how much would be the Loss% Decrease as a proportion of the absolute Gini increase for a given population bad rate and a given acceptance rate

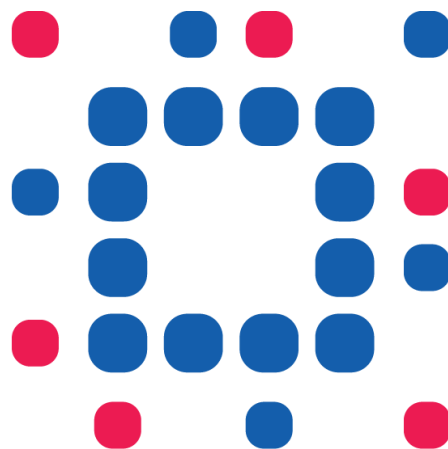


Acceptance Rate effect on the Relationship between Gini and %Loss Decrease

- Obviously the higher the Accept Rate the lower the effect of increasing the Gini Coefficient on Losses
- The Higher the Bad Rate of the total Population the smaller the effect (this difference is more pronounced at higher Accept Rates)



- The value of the function is 100% for Accept Rate of 60% and Bad Rate of 20% meaning that in this case if the Gini increases by 10% then Loss would decrease by the same %
- At an Accept Rate of 10% regardless of Portfolio Bad Rate a Gini increase of 10% would yield a Loss Decrease of 14% since the value of the function is 140%



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